

Audrey Gossard

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EDUCATION

PhD in Applied Mathematics 2023–2026

Laboratoire Jean Leray (Nantes), Inria Rennes

- Co-supervised by Annabelle Collin and Clair Poignard.
- *Thesis*: Asymptotic and homogenization methods for boundary value problems in bioimpedance microsystems.
- Asymptotic analysis, homogenization and numerical modeling of PDEs arising in bioimpedance, including an inverse problem based on experimental data.

Master 2 in Applied Mathematics 2022–2023

Sorbonne Université, Paris

- Mathematics for modeling in biology and medical sciences.

Engineering Degree 2019–2023

École Nationale des Ponts et Chaussées, Paris

- Department of Mathematics and Computer Science, specialization in mathematical modeling.

RESEARCH INTERNSHIPS

Modeling electroporation at the cellular scale for ventricular tachycardia 2023

Institut de Mathématiques de Bordeaux, Inria Bordeaux

6 months

- PDE-based modeling of a bioimpedance system for electroporation, involving bidomain formulations and transmission conditions.
- Finite element implementation and numerical simulation of the model.

Development of a reduced-order model for cardiovascular flows 2022

SISSA (Scuola Internazionale Superiore di Studi Avanzati), Trieste, Italy

6 months

- Development of a reduced-order model for cardiovascular flows using Proper Orthogonal Decomposition (POD).

Modeling the impact of climate change on aeronautics 2021

CERFACS, Toulouse

6 months

- Uncertainty quantification and sensitivity analysis for a thermodynamic model applied to aeronautical performance.
- Analysis of a global meteorological database.

TECHNICAL SKILLS

- **PDEs and modeling**: elliptic PDEs, boundary and transmission problems, asymptotic methods, homogenization.
- **Numerical methods**: finite element methods, convergence analysis, inverse problems and parameter estimation.
- **Scientific computing**: C++, Python, FreeFem++.
- **Interdisciplinary research**: mathematical modeling of biological systems and collaboration with experimental scientists.

CONFERENCES

17th World Congress on Computational Mechanics and 10th Congress on Computational Methods in Applied Sciences and Engineering 2026

Munich, Germany — Oral presentation

XXIX International Symposium on Bioelectrochemistry & Bioenergetics Bari, Italy — Oral presentation	2026
Congrès des Jeunes Chercheur·e·s en Mathématiques Appliquées Paris — Oral presentation	2026
12th Biennial Conference on Applied and Industrial Mathematics Gironde, France — Poster presentation — Best Poster Award	2025
5th World Congress on Electroporation Rome, Italy — Poster presentation	2024
MARGAUx PhD Days Pau, France — Poster presentation	2024

TEACHING

Numerical methods , L2, 36h. Nantes Université, Nantes	2026
Numerical analysis , L3, 16h. Nantes Université, Nantes	2026
Practical C++ programming for scientific computing , M1, 48h. ENSEIRB - Matméca, Bordeaux	2024
Modeling thermal systems with finite difference method , M1, 16h. ENSAM, Bordeaux	2024

LANGUAGES

French Native speaker
English Fluent
Italian Intermediate